Monopoly

Chapter 23

The Theory of Monopoly

- There is one seller
- The single seller sells a product for which there is no close substitute
- There are extremely high barriers to entry

Barriers To Entry

Legal Barriers: a Public Franchise is a right granted to a firm by government that permits the firm to provide a particular good or service and excludes all others from doing the same.

Natural Barriers:

- Economies of Scale: In some industries, low average total costs are only obtained through large scale production. If only one firm can survive in that industry, the firm is called a **Natural Monopoly**.
- Exclusive Ownership of a Necessary Resource: Existing firms may be protected from entry of new firms by the exclusive or near-exclusive ownership of a resource needed to enter the industry.

Government Monopolies Vs. Market Monopolies

Some economists use the term government monopoly to refer to monopolies that are legally protected from competition and the term market monopoly to refer to monopolies that are not legally protected from competition.

Monopoly Pricing and Output Decisions

- A monopolist is a price searcher; that is, it is a seller that has the ability to control to some degree the price of the product it sells.
- In the theory of monopoly, the monopoly firm is the industry and the industry is the monopoly firm. They are the same.

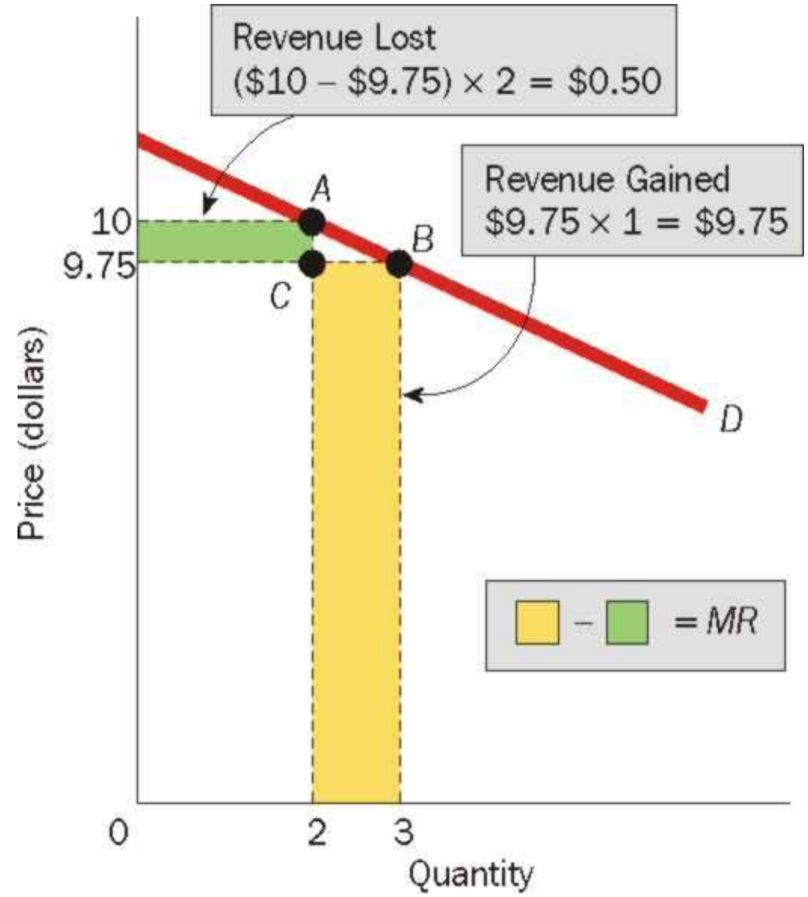
For Monopolists:

- Note that the price of the good being sold is greater than the marginal revenue. $P>MR$
- To sell an additional unit of a good (per time period), the monopolist must lower price.
- The monopolist gains and loses by lowering price.
- The gain equals the price of the product times one.
- The loss equals the difference between the new lower price and the old higher price times the units of output sold before the price was lowered.
- Marginal revenue can be defined as revenue gained minus revenue lost
- P=Revenue gained, MR=Revenue Gained – revenue lost, and revenue lost is >0. Therefore, $P>MR$

The Dual Effects of a Price Reduction on Total Revenue

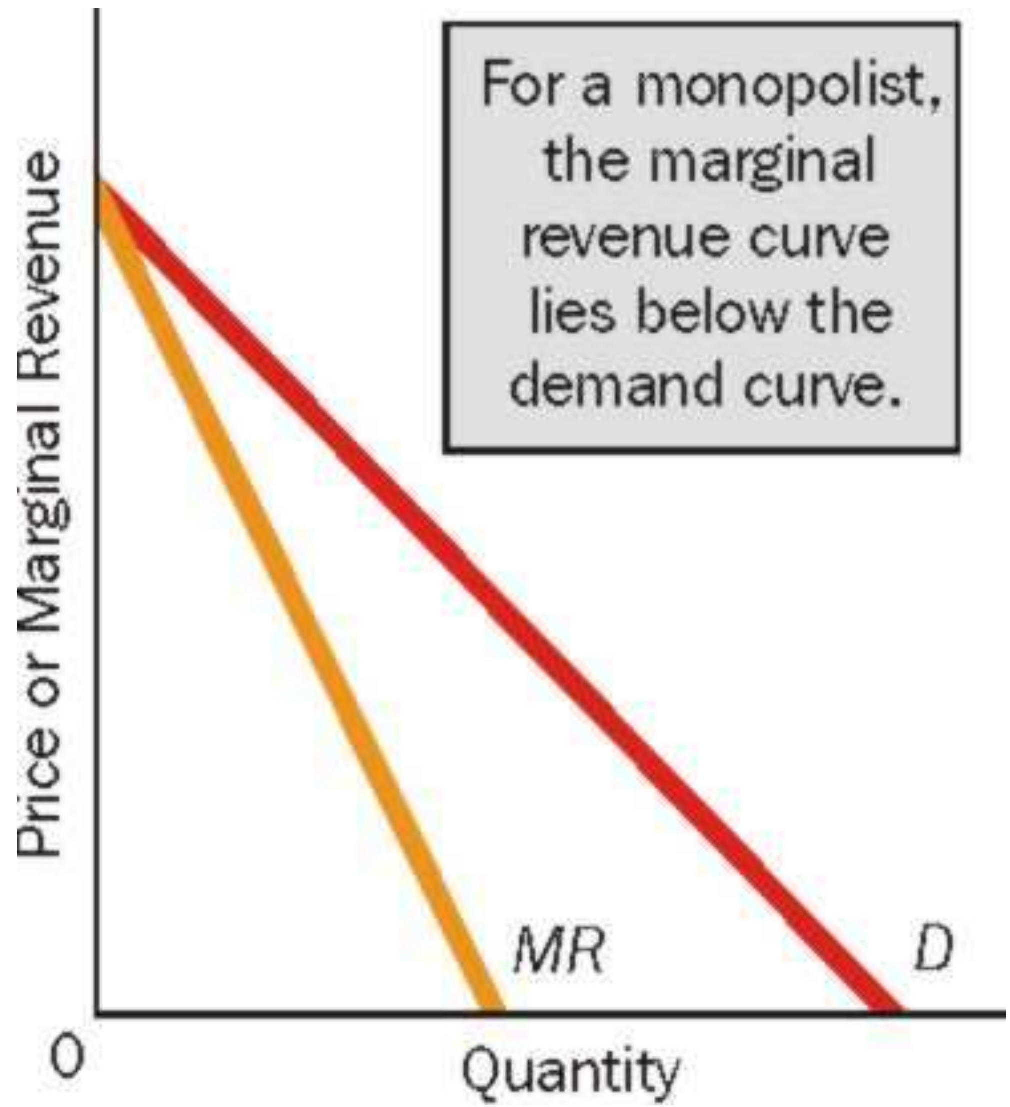
(1) *P*	(2) *Q*	(3) *TR*	(4) *MR*
$10.00	2	$20.00	$9.25
9.75	3	29.25	

To sell an additional unit of the good, a monopolist needs to lower price. This price reduction both gains revenue and loses revenue for the monopolist. In the exhibit, the revenue gained and revenue lost are shaded and labeled. Marginal revenue is equal to the larger shaded area minus the smaller.



Monopolist Demand and Marginal Revenue Curves

In monopoly, the firm’s demand curve is not the same as its marginal revenue curve. The monopolist’s demand curve lies above its marginal revenue curve.

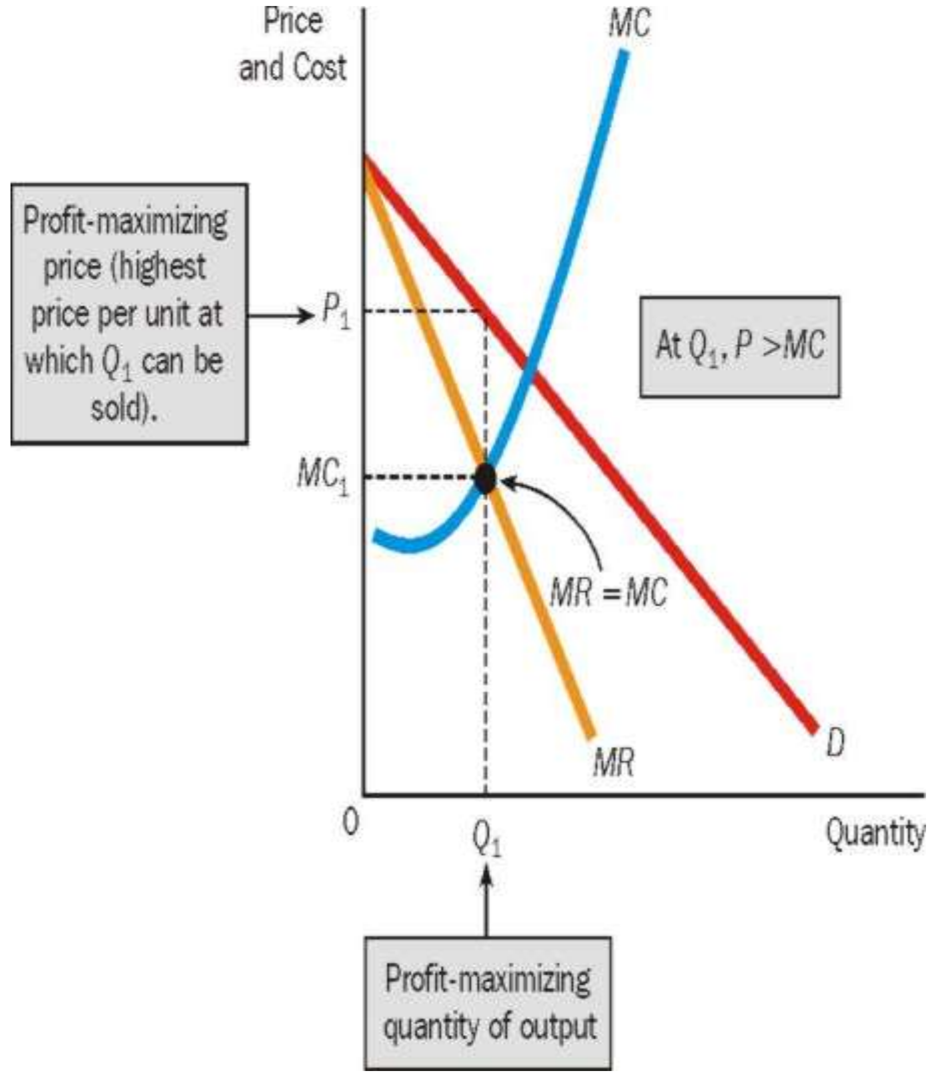


Maximizing Profit, Monopolist Style

- Maximizing revenues is the same as maximizing profits only when a firm has no variable costs. It is unlikely, though, that a firm will be without variable costs.
- The monopolist that seeks to maximize profits produces the quantity of output at which $MR=MC$ and charges the highest price per unit at which this quantity of output can be sold.

The Monopolist's Profit-Maximizing Price and Quantity of Output

The monopolist produces the quantity of output (Q_1) at which *MR*=*MC*, and charges the highest price per unit at which the quantity of output can be sold (P_1). Notice that at the profit maximizing quantity of output, price is greater than marginal cost, *P*>*MC*.



Competition Vs. Monopoly

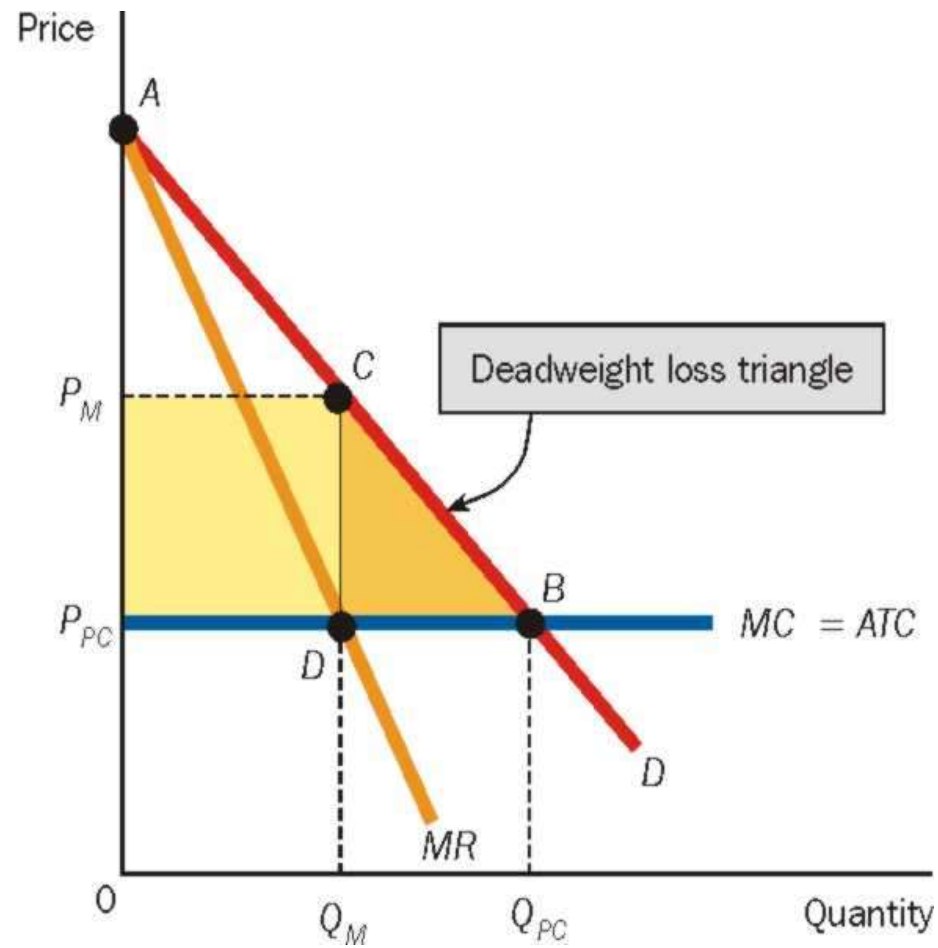
- For the perfectly competitive firm, $P=MR$; for the monopolist, $P>MR$. The perfectly competitive firm's demand curve is its marginal revenue curve; the monopolist's demand curve lies above its marginal revenue curve
- The perfectly competitive firm charges a price equal to marginal cost; the monopolist charges a price greater than marginal cost.
- A monopoly firm differs from a perfectly competitive firm in terms of how much consumers' surplus buyers receive.

The Case Against Monopoly

- The Deadweight Loss of Monopoly: Greater output is produced under perfect competition than under monopoly. The net value of the difference in these two output levels is said to be the deadweight loss of monopoly. This is the amount buyers value the additional output over and above the opportunity costs of producing the additional output.
- Rent Seeking: If firm A tries to get the government to transfer “income” or consumers’ surplus from buyers to itself it is undertaking a transfer seeking activity. In economics, these activities are usually called Rent Seeking.

Deadweight Loss and Rent Seeking as Costs of Monopoly

The monopolist produces QM, and the perfectly competitive firm produces the higher output level QPC. The deadweight loss of the monopoly is the triangle (DCB) between these two levels of output. Rent seeking is a socially wasteful activity because resources are expended to affect a transfer and not to produce goods and services.

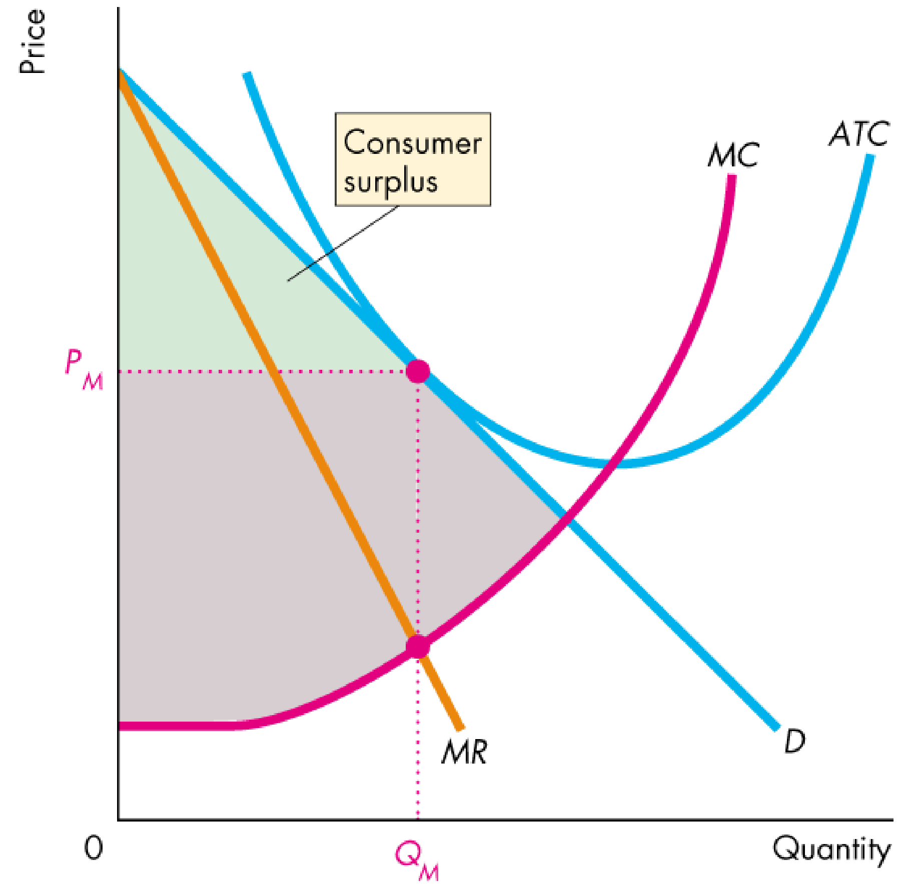


Rent Seeking

- The social cost of monopoly may exceed the deadweight loss through an activity called **rent seeking**, which is any attempt to capture consumer surplus, producer surplus, or economic profit.
- Rent seeking is not confined to a monopoly. There are two forms of rent seeking activity to pursue monopoly:
- Buy a monopoly—transfers rent to creator of monopoly.
- Create a monopoly—uses resources in political activity.

Rent Seeking

- Rent-Seeking Equilibrium
  - The resources used in rent seeking can exhaust the monopoly's economic profit and leave the monopoly owner with only normal profit.
  - Figure shows the normal profit that result from rent seeking.



X-Inefficiency

Refers to The increase in costs when monopolists are operating at higher than lowest possible costs, and to the organizational slack that is directly tied to this.

Price Discrimination

- Price discrimination occurs when the seller charges different prices for the product it sells, and the price differences do not reflect costs.
- **Perfect Price Discrimination**: sells each unit separately and charges the highest price each consumer would be willing to pay for the product.
- **Second Degree Discrimination**: it charges a uniform price per unit for one specific quantity, a lower price for an additional quantity, and so on.
- **Third Degree Discrimination**: it charges a different price in different markets or charges a different price to different segments of the buying population

Conditions of Price Discrimination

Conditions of Price Discrimination:

- – The seller must exercise some control over price; it must be a price searcher.
- – The seller must be able to distinguish among buyers who would be willing to pay different prices.
- – It must be impossible or too costly for one buyer to resell the good at other buyers. The possibility of arbitrage, or "buying low and selling high" must not exist.

Price Discrimination

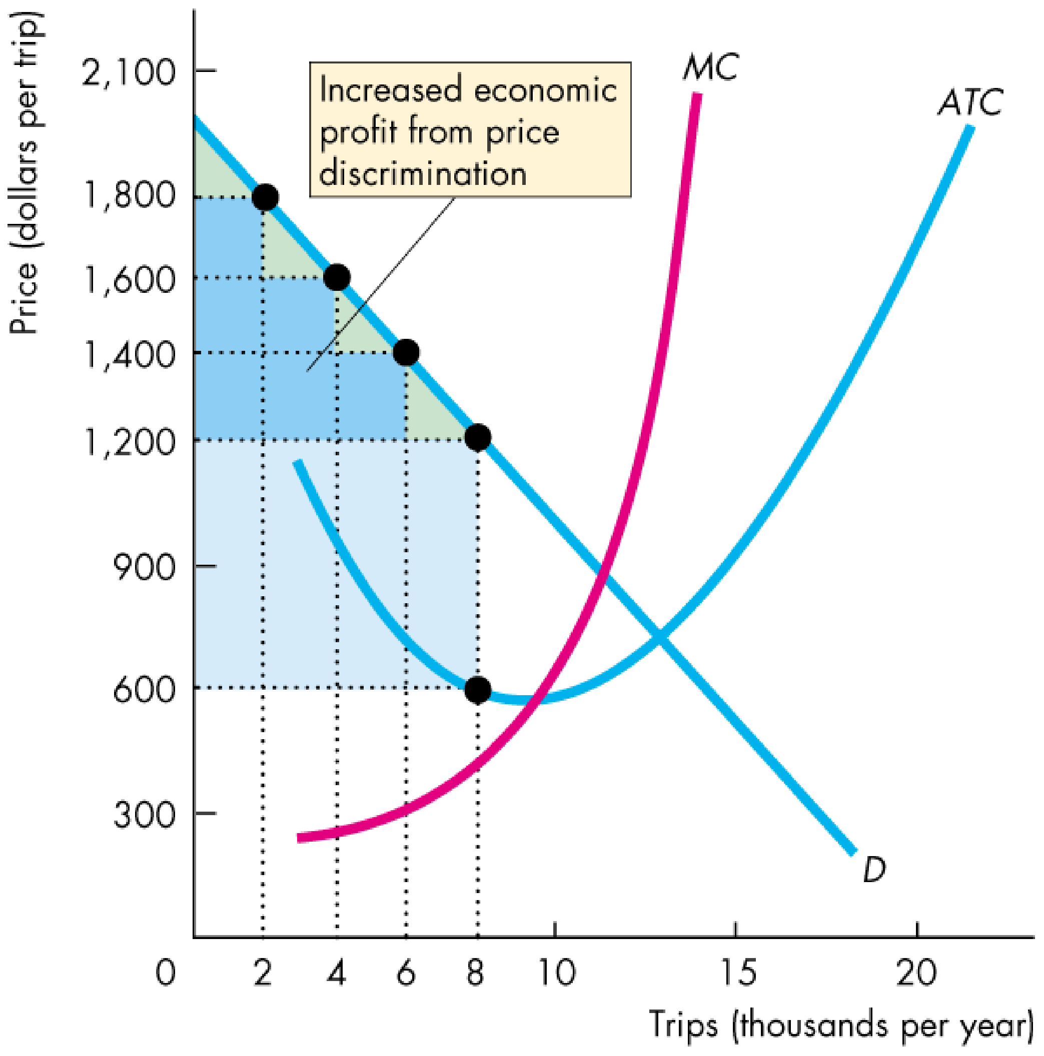
- The perfectly price-discriminating monopolist tries to get the highest price for each customer, irrespective of what other customers pay.
- **Perfect price discrimination** is a price structure in which the seller charges the highest price that each consumer is willing to pay for the product rather than go without it.

The perfectly price discriminating monopolist and the perfectly competitive firm both exhibit **resource allocative efficiency**.

Difference between single-price monopoly, perfectly price discriminating monopoly and perfectly competitive firm.

By price discriminating, the firm can increase its profit.

In doing so, it converts **consumer surplus into economic profit.**



Second Degree and Third Degree Price Discrimination

- 2nd degree price discrimination is a price structure in which the seller charges a uniform price per unit for one specific quantity, a lower price for an additional quantity, and so on. Quantity discount is an example of this.
- 3rd degree price discrimination is a price structure in which the seller charges different prices in different markets or charges different prices to various segments of the buying population.

In 3rd degree price discrimination monopolist will charge relatively higher price where the demand is less elastic. In the following figure monopolist is charging higher price in market B. Shaded area shows the profit in each market. MR of the last unit sold in each market is the same.

So, $MC = MR_A = MR_B$

3rd degree price discrimination